A Resident-Streaming Roofline Model for Compute-in-Memory

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***Abstract*—From CPUs to GPUs, the Roofline model has become a textbook tool for revealing a platform's performance ceiling and how well a workload matches it. Compute-in-memory (CIM), however, falls outside this picture because it violates the core assumption of classical model. Here, we present a resident-streaming Roofline model for CIM, in which streaming-activation and resident-weight throughputs replace compute throughput and memory bandwidth, respectively. This follows from a symmetry we observe: conventional processors pair separated hardware with homogeneous data, whereas CIM pairs unified hardware with heterogeneous data—operands residing in the array and operands streaming through it. Using NeuroSim with literature-extracted parameters, we evaluate various CIM technologies on this Roofline and quantify the demands of mainstream large language models. With similar periphery, the streaming throughputs of these technologies differ by a factor of only about 10^2, whereas their resident throughputs differ by over 10^4. The memory thus decides mainly how much reuse CIM needs, from about 2–10 input vectors per weight load on SRAM and eDRAM to about 4×10^4 on NOR flash, making the reuse a workload offers an explicit criterion for choosing among technologies.**

***Index Terms*—Compute-in-memory, roofline model, streaming intensity, data reuse, large language models.**

I. Introduction

The textbook Roofline model [1] relates hardware capability to workload demand through the Arithmetic Intensity (AI), and exposes the bottleneck arising from a mismatch between soft/hardware. It describes the performance on computation and on data transfer by the compute capability π and the memory bandwidth β, respectively. AI represents the number of operations a workload supports per byte of data transferred; comparing it with the hardware ridge point $\mathrm{AI}^* = \pi/\beta$ distinguishes the memory-bound region from the compute-bound region and provides guidance for optimization [2], [3].

The validity of this model rests on an assumption implicit in the von Neumann architecture: memory and compute are separate units [4]. Fig. 1(a) shows the familiar picture. Storage and computation are carried out by different hardware, and although activations A and weights W play different roles in the algorithm, the hardware treats both simply as operands delivered to the compute units: neither has a distinct identity, and the two are interchangeable.

However, this assumption no longer holds in Compute-in-Memory (CIM), a non-von Neumann architecture [5]. In Fig. 1(b), the CIM macro that stores W also calculates A, so the two hardware units are unified [6]. The ***weights*** take part in computation as ***resident state***, while each new ***activation*** enters through the read path and ***streams across*** the macro, where it is calculated against that state. CIM turns one class of operands, the weights, into stored state at the very site of calculation, while the other class, the activations, flows through the macro as inputs served by that state. The separate hardware units of memory and compute in CPU/GPU disappear, but new roles of streaming and resident operands emerge in CIM.

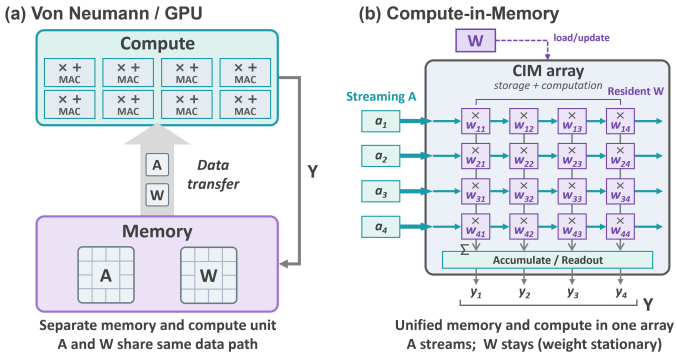


Fig. 1. Hardware schematics: (a) CPU/GPU: separated hardware with homogeneous data; (b) CIM: unified hardware with heterogeneous data.

This change calls for a new Roofline model that rethinks how compute capability and data transfer relate in CIM. The need has two roots. First, the hardware fusion of the CIM architecture dissolves the concept of compute capability π: the capability of a CIM macro to complete multiply-accumulate (MAC) is, in essence, the "transfer rate" of A. Second, the split of operand roles creates two distinct throughputs: weights enter through the write path, which erases, programs, or refreshes the cells, whereas activations enter through the read path, which applies them as inputs and accumulates their products with the weights as outputs. Hence, whereas AI captures the GPU bottleneck as an imbalance between computation and memory access, the CIM bottleneck is the imbalance between these two operand throughputs. Prior studies have recognized this problem [7] and proposed several Roofline analyses for CIM [2], [8]–[14], yet none of them reveals this difference.

Therefore, we propose a resident-streaming Roofline model for CIM. This model separately counts the input bytes Q_S consumed by streaming calculation and the write bytes Q_R that update or reload resident state, and defines the Streaming Intensity $\mathrm{SI} = Q_S/Q_R$ as the streaming demand supported per byte of resident write. Q_S is served by the streaming throughput ρ and Q_R by the resident throughput τ, so the model keeps the form of the classical Roofline, with ρ, τ, and SI taking the places of π, β, and AI. By distinguishing the two operand roles, streaming A and resident W, the model quantifies the imbalance between resident updates and streaming calculation: its ridge $\mathrm{SI}^* = \rho/\tau$ marks where the bottleneck shifts from resident updates to streaming calculation.

Building on this model, we carry out a quantitative analysis from two sides: hardware capability and algorithm demand. On the hardware side, we characterize the streaming and resident throughputs of ten CIM technologies through NeuroSim simulations calibrated with parameters extracted from the literature.

On the algorithm side, we examine the matrix multiplications (GEMM) in mainstream large language models (LLM) and characterize the demands posed by model weights and by key-value (KV) caches. The analysis shows that, with similar periphery, the streaming throughputs of the ten technologies differ by a factor of only about 10^2, whereas their resident throughputs differ by over 10^4. Hence the ridge SI* is limited by the write mechanism of each technology. Comparing the SI of a workload with SI* thus shows which technologies can run it at their streaming ceiling, providing clear guidance for matching CIM technologies to workloads.

II. Proposed Resident-Streaming Roofline

A. Workload Demands and Hardware Throughputs

The classical Roofline describes a workload by its operation count (OPs) and data-transfer volume (Bytes). Following the operand roles in Fig. 1(b), we instead describe a CIM workload through its streaming inputs and resident writes. We count both over an explicit *workload window*, which specifies the work under analysis and the resident state at its start.

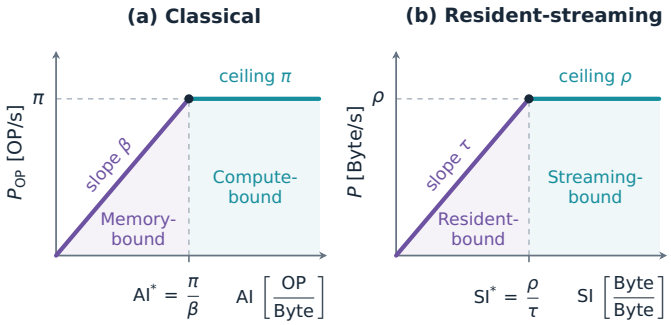
We measure streaming calculation by the logical input bytes evaluated using resident state, with each evaluation (*multiplication*) producing the specified outputs (*partial sum*). The streaming demand Q_S is the number of such bytes served at the chosen logical input boundary; it thus represents completed calculation work rather than input/output traffic alone, and it is directly comparable with resident writes. In Fig. 1(b), Q_S counts logical input bytes entering the CIM I/O and *broadcast* along wordlines (WLs), without counting broadcast copies. The resident demand Q_R is the number of logical bytes written within the window to load or update the resident state that this work requires. An initial load completed before the window is excluded; all updates and reloads within the window are counted. The resident operand may be a static model weight or the dynamic KV cache of earlier tokens.

We characterize two hardware services, streaming-input calculation and resident-state loading, by their throughputs. The streaming throughput ρ is the maximum rate, in logical input bytes per second, at which the hardware completes streaming calculations and produces the specified outputs. The resident throughput τ is the maximum rate at which logical resident bytes become ready for calculation. Both rates refer to the CIM hardware under analysis, with bytes counted at the same interfaces and operand precisions used for Q_S and Q_R, respectively. We define Streaming Intensity and average streaming-input throughput as

$$\mathrm{SI} = \frac{Q_S}{Q_R}, \qquad P = \frac{Q_S}{T} \quad [\mathrm{Byte/s}], \tag{1}$$

where T is the window duration, P is the average streaming-input throughput over the window, and SI is the streaming demand per byte of resident write.

The workload's SI describes the relative demand for streaming calculation and resident writes, while ρ and τ describe how quickly the hardware can perform each. The throughput bound therefore follows from the time required to complete these two kinds of work.



Aspect	GPU (a)	CIM (b)
Decomposition	Hardware resources	Data roles
Workload	OP, Byte	Q_S, Q_R
Service rates	π [OP/s], β [Byte/s]	ρ, τ [Byte/s]
Intensity	AI = OP/Byte	SI = Q_S/Q_R
Vertical axis	P_{OP} = OP/T	P = Q_S/T
Upper bound	$P_{OP} \le \min(\pi, \beta\,\mathrm{AI})$	$P \le \min(\rho, \tau\,\mathrm{SI})$
Left regime	Memory-bound	Resident-bound
Right regime	Compute-bound	Streaming-bound
Ridge	AI* = π/β	SI* = ρ/τ

Fig. 2. Classical GPU [1] and resident-streaming CIM Rooflines with their accounting correspondence. Panels use independent linear scales; curves are upper bounds. OP and Byte denote operation and transferred-byte counts.

B. Theoretical Model and Correspondence

Fig. 2 compares the proposed resident-streaming Roofline with the classical model. Streaming calculation and resident writes require at least Q_S/ρ and Q_R/τ of service time, respectively. The execution must satisfy both constraints, giving

$$P \le \min(\rho, \tau \cdot \mathrm{SI})\,. \tag{2}$$

The two service requirements balance at the hardware ridge $\mathrm{SI}^* = \rho/\tau$. A workload is resident-bound on this hardware when $\mathrm{SI} < \mathrm{SI}^*$ and streaming-bound when $\mathrm{SI} > \mathrm{SI}^*$.

The same bound can be expressed in the familiar GPU-style operation and byte rates. For a window containing O useful operations, the equivalent compute throughput π_{eq} and aggregate bandwidth β_{eq} of CIM are

$$\pi_{\mathrm{eq}} = \frac{O}{Q_S} \cdot \min(\rho, \tau \cdot \mathrm{SI})\,, \tag{3}$$

$$\beta_{\mathrm{eq}} = \left(1 + \frac{Q_R}{Q_S}\right) \cdot \min(\rho, \tau \cdot \mathrm{SI})\,. \tag{4}$$

These are workload-specific projections of the same upper bound: β_{eq} counts the combined logical volume $Q_S + Q_R$, rather than traffic through a GPU memory interface.

Whether a hardware metric is workload-independent depends on the coordinates used to describe it. In GPU, compute

TABLE I
Streaming Intensity after one complete weight load

Weight shape	Reuse count U					
$N \times K$	128	1K	16K	128K	1M	$\to \infty$
128×128	1.0	8.0	128.0	1024.0	8192.0	∞
1024×1024	0.125	1.0	16.0	128.0	1024.0	∞
4096×4096	0.0313	0.250	4.0	32.0	256.0	∞
1024×4096	0.125	1.0	16.0	128.0	1024.0	∞
4096×1024	0.0313	0.250	4.0	32.0	256.0	∞

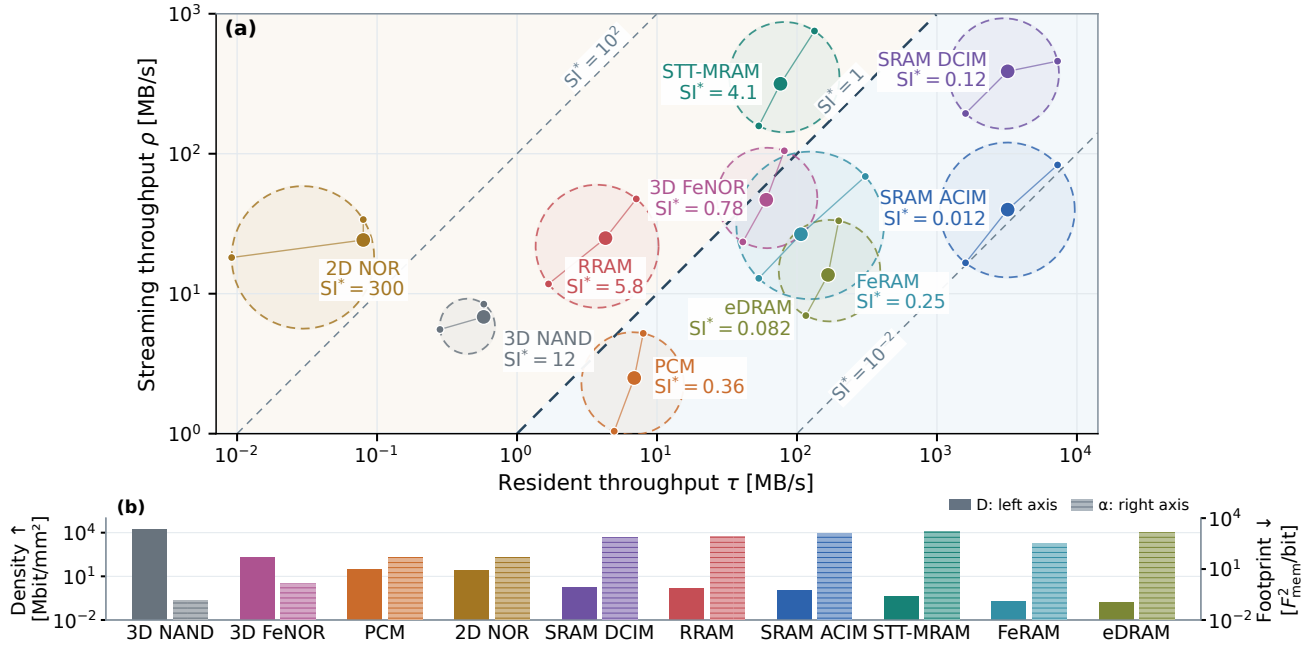


Fig. 3. Throughput and storage compactness of ten CIM technologies. (a) Streaming throughput ρ and resident throughput τ estimated with NeuroSim using literature-extracted parameters. The wider variation in τ drives balance intensities $\mathrm{SI}^* = \rho/\tau$ and reuse requirements. Large dots mark reference configurations; smaller dots and circles show scenario spread. (b) Binary storage density and normalized bit footprint from independent cell/tile geometry references.

and memory units provide distinct services, matched to operation counts and data-transfer volumes. In CIM, storage and computation are combined, but streaming calculation and resident-state loading remain distinct services, matched to streaming-input and resident-write bytes. Each model therefore decomposes hardware services and workload demands along the same dimension—by hardware function for GPUs and by operand role for CIM. This alignment gives each architecture its native capabilities, (π, β) or (ρ, τ), which are fixed for a given configuration. Translating between the two descriptions requires additional information: how many operations the inputs support and how much resident writing is needed. The hardware remains unchanged, but its projected metrics become workload-dependent because the relationship between operations and operand bytes is determined by the workload.

C. Analytical Case Studies

Consider an illustrative 28 nm SRAM ACIM macro serving a logical 128×128 INT8 matrix, with preloaded weights and an assumed input-vector service interval $\Delta_S = 10\,\mathrm{ns}$. Each matrix-vector multiplication consumes 128 input bytes and performs 2×128^2 useful operations, counting two operations per MAC. Thus $Q_R = 0$ and $O/Q_S = 256\,\mathrm{OP/Byte}$, giving

$$\begin{aligned} \beta_{\mathrm{eq}} = \rho &= \frac{128\,\mathrm{Byte}}{10\,\mathrm{ns}} = 12.8\,\mathrm{GB/s}, \\ \pi_{\mathrm{eq}} &= (256\,\mathrm{OP/Byte}) \cdot \rho \approx 3.28\,\mathrm{TOPS}. \end{aligned} \tag{5}$$

The familiar TOPS value is therefore the streaming ceiling expressed in operation units.

We next evaluate SI for GEMM workloads. For a logical matrix $W[N, K]$ with N outputs and K inputs, let U be the number of input vectors served after one complete load, including the first use and allowing reuse across batches or requests. With one byte per input and weight element, $Q_S = UK$, $Q_R = NK$, and $\mathrm{SI} = U/N$. For full-load time T_R and vector service interval Δ_S, the corresponding balance occurs at $U^* = T_R/\Delta_S = N\mathrm{SI}^*$. Table I lists the resulting SI for five matrix shapes and representative reuse counts.

III. Hardware and Workload Characterization

Fig. 3(a) presents ρ and τ estimated using NeuroSim [15], [16] with literature-extracted parameters for SRAM DCIM [17]–[19], SRAM ACIM [7], [20], [21], 2D NOR [22]–[24], 3D NAND [25]–[27], RRAM [28]–[30], STT-MRAM [31]–[33], PCM [34]–[36], FeRAM [37]–[39], eDRAM [40]–[42], and 3D FeNOR [43]–[45]. Common peripheral assumptions [46]–[48] and logical-byte accounting provide a consistent comparison while retaining native array sizes and update organizations. These are macro-level calculation and update rates, including required erase, verification and refresh costs, rather than the aggregate I/O bandwidth of a multi-bank system. Three paired scenarios per configuration reflect literature-informed parameter ranges. In log space, each circle passes through the two alternative points with its center nearest the reference point; it visualizes scenario spread rather than a capability boundary.

Points toward the upper right provide higher throughputs for both services, while the diagonal guides mark constant SI^*. Above $\rho = \tau$ (yellow), resident updates are slower per byte than streaming calculation; below it (blue), the reverse holds. 3D FeNOR lies close to this diagonal, with relatively balanced streaming and resident throughputs.

Within each scenario, ρ varies across technologies by about 10^2, but τ by over 10^4: similar peripheral conditions constrain streaming calculation, whereas write mechanisms and update parallelism differ greatly. Reference SRAM ACIM and NOR illustrate this distinction: their streaming rates are about 40 and 24 MB/s, but resident rates are 3200 and 0.080 MB/s,

TABLE II
STREAMING INTENSITIES OF REPRESENTATIVE INFERENCE MATRIX STAGES

	Weight projections						Attention					
	QKV (U)			FFN/MoE (U)			Prefill (L)			Decode (L)		
Model	1K	128K	1M	16	1K	16K	1K	8K	64K	1K	8K	64K
Qwen3.5-2B	0.200	25.6	204.8	0.00347	0.222	3.6	6.0	34.0	258.0	10.0	66.0	514.0
Ministral 3 8B	0.167	21.3	170.7	0.00167	0.107	1.7	10.0	66.0	514.0	18.0	130.0	1026.0
Qwen3.6-35B-A3B	0.111	14.2	113.8	0.0130	0.833	13.3	12.0	68.0	516.0	20.0	132.0	1028.0
Hy3 (295B)	0.100	12.8	102.4	0.00477	0.306	4.9	20.0	132.0	1028.0	36.0	260.0	2052.0
Ling-1T (1000B)	0.100	12.8	102.4	0.00326	0.208	3.3	20.0	132.0	1028.0	36.0	260.0	2052.0
MiMo-V2.5-Pro (1020B)	0.0377	4.8	38.6	0.00347	0.222	3.6	35.2	214.4	1648.0	60.8	419.2	3286.4

Entries are SI (one byte/element; $1\text{K} = 1024$, $1\text{M} = 1024^2$). QKV includes any gate but excludes O; FFN/MoE covers one dense FFN or routed expert. Prefill starts with an empty KV cache; decode appends one token to reach visible length L. Ling-1T uses YaRN context extension at 64K.

respectively, leading to very different reuse requirements. Storage density, however, also matters in technology selection. Fig. 3(b) adds independent binary cell/tile density and normalized footprint estimates from reported geometry and conditional models [49]–[51], including a 500-layer projection of NAND geometry [25]. Compact storage can make slower-updating technologies attractive for retaining large resident states, so the upper-right position alone is not a sufficient selection criterion.

Table II reports SI for 72 matrix-stage operating points derived from six fixed model configurations [52]–[57], using one byte per element. Weight projections and FFN/MoE follow the one-load accounting of Section II-C, counting shared branch inputs once and each subsequent stage's new input separately. For causal grouped-query attention (GQA), H_q and H_{kv} denote query and KV head counts, and d_{QK} and d_V the query/key and value dimensions. Counting queries and attention coefficients as streaming inputs and writing each token's KV once, prefill from an empty cache to length L gives

$$Q_S = LH_q[d_{\text{QK}}+(L+1)/2], \quad Q_R = LH_{\text{kv}}(d_{\text{QK}}+d_V). \quad (6)$$

Decode appends one token to reach L: omit the leading L in both counts and replace $(L+1)/2$ by L in Q_S.

The increases in SI in Table II have different origins. For weight stages, larger U spreads a fixed load over more input vectors, reducing resident writes per use. For attention decode, larger L increases streaming work per token, while the logical KV write volume per token remains fixed. The resulting higher SI therefore accompanies more calculation and a larger cache: sustaining the same token rate requires greater streaming throughput and storage capacity.

IV. WORKLOAD MATCHING AND DESIGN IMPLICATIONS

Fig. 4 compares six Qwen3.6-35B-A3B operating points from Table II with the native hardware balance intensities shown in Fig. 3(a). The SI–SI* comparison identifies the tighter service requirement for each stage and window. Increasing weight reuse can shift the bottleneck from resident updates to streaming calculation, while near-ridge matches remain sensitive to hardware conditions. These are reference comparisons; deployment requires aligned accounting for input sharing, replay and resident updates.

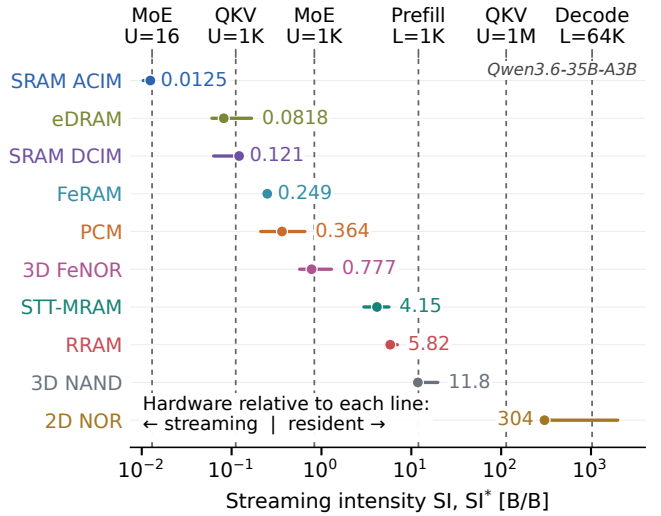


Fig. 4. CIM–workload matching. Dots and ranges show hardware balance intensities SI*; dashed lines show workload SI. Technologies left of each line are streaming-bound; those to its right are resident-bound.

Technology selection must consider absolute throughput and storage compactness, rather than SI* alone: a low SI* may reflect slow streaming calculation rather than fast resident updates. For resident-bound work, faster updates (program/erase) or greater SI through fewer reloads raise the bound, whereas improving ρ alone does not. For streaming-bound work, the priority is to reduce the delays limiting evaluation throughput, such as ADC conversion time, adder-tree latency, or array/interconnect RC delay. Compact storage (3D stacking) can help sustain weight reuse by keeping more state resident, but the benefit depends on usable capacity and the replacement policy. Growing KV caches also require an update organization suited to incremental writes.

V. CONCLUSION

Outline. Close with the role-based connection between in-window resident updates and streaming evaluation established by (2). Use Fig. 4 to emphasize the practical outcome: hardware service asymmetry becomes an explicit reuse requirement for residency and update decisions.

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